

Q-1 List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Main Components Inside a Desktop PC

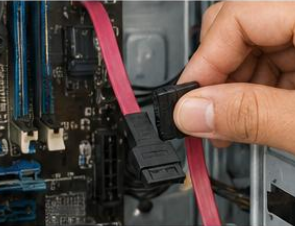
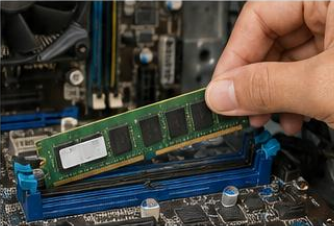
1. **Motherboard**
The main circuit board that connects and allows communication between all computer components.
2. **CPU (Central Processing Unit)**
The brain of the computer that processes instructions and performs calculations.
3. **CPU Cooler (Heatsink and Fan)**
Cools the CPU by dissipating heat and preventing overheating.
4. **RAM (Random Access Memory)**
Temporary memory used by the CPU to store data while programs are running.
5. **Storage Drive (HDD/SSD)**
Stores the operating system, applications, and user files permanently.
6. **Graphics Card (GPU)**
Processes graphics and video output for gaming, video editing, and other graphics-intensive tasks.
7. **Power Supply Unit (PSU)**
Converts AC power into DC power and supplies electricity to all internal components.
8. **Case Fans**
Improve airflow inside the cabinet and help keep components cool.
9. **Optical Drive (Optional)**
Reads and writes CDs, DVDs, or Blu-ray discs.
10. **SATA Data and Power Cables**
Connect storage devices to the motherboard and power supply.
11. **PCIe Expansion Cards (Optional)**
Add extra features such as Wi-Fi, sound, or additional USB ports.
12. **CMOS Battery**
Maintains BIOS settings and keeps the system clock running when the PC is powered off.

Q-2 Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

DISASSEMBLY OF A DESKTOP PC – STEP BY STEP

Tools used: Phillips screwdriver, flathead screwdriver

1 Unplug All Cables  Turn off the PC and unplug the power cable and all other connected cables.	2 Remove Side Panel  Remove the screws from the back of the case and slide off the side panel.	3 Inside View  This is how the inside of the PC looks. Identify the components before removing.
4 Disconnect Power Cables  Disconnect the 24-pin motherboard power cable and 4/8-pin CPU power cable.	5 Disconnect Data Cables  Disconnect SATA data cables from HDD/SSD and any other devices.	6 Remove RAM  Press the clips on both sides of the RAM stick and gently remove it from the slot.
7 Remove Storage (HDD/SSD)  Remove the screws holding the HDD/SSD in place and take it out. (Disconnect cables already.)	8 Remove Expansion Cards (if any)  Remove the screw from the bracket, press the clip (if any) and gently pull the card out of the PCIe slot.	9 Remove CPU Cooler  Unplug the CPU fan cable and unscrew the cooler. Then gently lift it off.
10 Remove Motherboard Screws  Remove all screws holding the motherboard in place.	11 Remove Motherboard  Gently lift the motherboard and take it out of the case.	12 All Components Disassembled  All major components have been removed successfully.

Q-3 Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Reassembling a Desktop PC

Step 1: Install the Motherboard

Explanation: Place the motherboard inside the cabinet and secure it with screws. The motherboard must be installed first because all other components connect to it.

Step 2: Install the CPU

Explanation: Carefully align the CPU with the socket and lock it into place. Proper installation ensures the processor can communicate with the motherboard.

Step 3: Install the CPU Cooler

Explanation: Attach the heatsink and fan to the CPU and connect the fan cable to the CPU_FAN header. This prevents the CPU from overheating.

Step 4: Install the RAM

Explanation: Insert the RAM module into the correct memory slot until the locking clips click into place. RAM is required for the system to run.

Step 5: Install the Storage Drive (HDD/SSD)

Explanation: Mount the HDD or SSD in its bay and connect the SATA data cable to the motherboard and the SATA power cable from the PSU.

Step 6: Install the Graphics Card (GPU)

Explanation: Insert the graphics card into the PCIe slot and secure it with screws. Connect the PCIe power cable if required.

Step 7: Connect the Power Supply Cables

Explanation: Connect the 24-pin motherboard power cable, 8-pin CPU power cable, GPU power cable (if needed), and SATA power cables to the storage drives.

Step 8: Connect Front Panel Cables

Explanation: Connect the power switch, reset switch, power LED, HDD LED, USB, and audio cables to the motherboard.

Step 9: Connect Case Fans

Explanation: Plug the case fan cables into the motherboard or power supply to ensure proper airflow and cooling.

Step 10: Close the Cabinet and Test

Explanation: Reattach the side panel, tighten the screws, reconnect the power cable, and turn on the PC to verify that all components are working correctly.

Q-4 Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

- [PC Assembly/Disassembly Checklist](#)

Safety Precautions

☐ **Turn off the computer and unplug the power cable**

Why it's important: Prevents electric shock and protects components from electrical damage.

☐ **Press the power button after unplugging**

Why it's important: Discharges any remaining electricity stored in the system.

☐ **Wear an anti-static wrist strap**

Why it's important: Prevents electrostatic discharge (ESD), which can damage sensitive components.

☐ **Work on a clean, flat, non-conductive surface**

Why it's important: Reduces the risk of short circuits and keeps components safe.

☐ **Handle components by their edges**

Why it's important: Prevents damage to chips, pins, and electrical contacts.

☐ **Keep food and liquids away from the work area**

Why it's important: Prevents accidental spills that could damage the computer.

☐ **Store screws in a small container**

Why it's important: Keeps screws organized and prevents them from getting lost.

☐ **Follow the motherboard or PC manual**

Why it's important: Ensures components are installed and removed correctly.

- Tools Needed

- ☐ **Phillips-head Screwdriver**

Why it's important: Used to remove and tighten most PC screws.

- ☐ **Anti-static Wrist Strap**

Why it's important: Protects computer components from static electricity.

- ☐ **Small Container or Magnetic Tray**

Why it's important: Keeps screws and small parts organized during assembly.

- ☐ **Soft Brush or Compressed Air**

Why it's important: Removes dust from components without causing damage.

- ☐ **Cable Ties (Zip Ties)**

Why it's important: Organizes cables and improves airflow inside the PC.

- ☐ **Flashlight (Optional)**

Why it's important: Helps you clearly see small connectors and screws inside the cabinet.

- ☐ **Thermal Paste (If Removing the CPU Cooler)**

Why it's important: Ensures proper heat transfer between the CPU and the heatsink after reinstallation.